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EXPERIMENT – 30

Analysis of Cellular Frequency Planning and Handover Performance under User Mobility Conditions Using MATLAB

PROGRAM:

```
clc;
clear;
close all;
% Cell Centers (3 Cells)
cellPos = [0 500 1000];
% User Movement
user = 0:10:1000;
% Nearest Cell Selection (Handover)
handover = zeros(size(user));
for i = 1:length(user)
[~,handover(i)] = min(abs(user(i)-cellPos));
end
% Display
disp('User Position Connected Cell')
disp([user' handover'])
% Plot
figure;
subplot(2,1,1)
plot(user,handover,'b','LineWidth',2)
title('Handover Performance')
xlabel('User Position (m)')
ylabel('Cell Number')
grid on
```

```
subplot(2,1,2)
stem(cellPos,[1 2 3],'filled')
hold on
plot(user,ones(size(user))*2,'r--')
title('Cellular Frequency Planning')
xlabel('Distance (m)')
ylabel('Cell')
grid on
```

OUTPUT:

